Team Contributions: POC Software Engineering

Team 11, technically functional
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This document summarizes the contributions of each team member up to the POC Demo. The time period of interest is the time between the beginning of the term and the POC demo.

1 Demo Plans

For the POC demo we plan to showcase the two major risks of our project which were identified in the development plan document. A short section has been copied for reference.

A few major risks in our proof of concept demonstration are:

- Risks associated with measuring a 3D motion with a 2D view/plane. Currently the scope of the project includes patients with exercises to the knee joint due to complexities with joints that are omni-directional (i.e. shoulder, ankle, etc.).
  - To address this, the team will test how a single camera view is sufficient for recording the movement. This will be tested primarily on the sagittal view and compared with adding frontal and transverse views. The accuracy of video analysis will be measured by obtaining the results from each view and comparing the difference in guided prompts and verifying if they can adhere to the requirements of the exercise (i.e. does a single view yield false positives? Does having two views change the accuracy achieved?).
- The second is being able to measure metrics related to stretching and past history. This was identified as a key risk since range of motion is a key metric in physiotherapy and measuring it over time is a good indicator of exercise efficacy. Furthermore, while talking with the advisor for

the project it was apparent that patients are not "pushing themselves" mobilizing their knee as much due to the post-operative condition.

  – This can be measured by testing the system to see if it can recognize any marginal increase within the exercise (i.e. incremental changes in the angle of knee flexion). This can help indicate the limitations of ranges of motion and measure exercise duration.

2 Team Meeting Attendance

Student	**Meetings**
Total	15
Matthew	15
Cieran	13
Vaisnavi	14
Maham	14
Eman	14

The above table does not include attendance for TA meetings during tutorial as well as smaller meetings for 'sub team work' after the VnV since not all team member's attendance was needed.

3 Supervisor/Stakeholder Meeting Attendance

Supervisor's Name: Kevin Yu

Student	**Meetings**
Total	2
Matthew	2
Cieran	2
Vaisnavi	2
Maham	2
Eman	2

Currently the total number of (virtually) meetings have been 2 but we have periodically provided Kevin with updates on our project and we are progressing.

4 Lecture Attendance

Student	Lectures
Total	6
Matthew	6
Cieran	4
Vaisnavi	4
Maham	5
Eman	2

This table takes into account the software lectures from our initial group forming on Sep 15. This also does not take into account the guest lecture on Sep 29.

5 TA Document Discussion Attendance

TA's Name: Rashad Bhuiyan

Student	Meetings
Total	3
Matthew	3
Cieran	3
Vaisnavi	3
Maham	3
Eman	2

These meeting were for the Problem Statement and Development Plan, SRS and HA, and the VnV Plan.

6 Commits

Student	Commits	Percent
Total	174	100%
Matthew	45	25.8%
Cieran	53	30.4%
Vaisnavi	27	15.5%
Maham	45	25.8%
Eman	4	2.3%

** add justification here, I know for Vaisnavi, a few commits werent tracked at the beginning of the project due to her account settings?

[If needed, an explanation for the counts can be provided here. For instance, if a team member has more commits to unmerged branches, these numbers can be provided here. If multiple people contribute to a commit, git allows for multi-author commits. —SS]

7 Issue Tracker

Student	Authored (O+C)	Assigned (C only)
Matthew	73	13
Cieran	9	5
Vaisnavi	2	8
Maham	5	14
Eman	0	4

A majority of the assigned column is not done properly and mainly encapsulates the SRS and HA sections as that was the only deliverable that created issues for each section and were assigned accordingly but that was based on a different SRS template and doesn't capture the template that we used (Volere).

8 CICD

[Say how CICD will be used in your project —SS]

9 Team Charter Trigger Items

[Provide a summary of the quantified triggers identified in the team's charter. —SS]

[Provide a list of any violations of the triggers. If the team wishes, the violations can be summarized on aggregate, instead of naming specific team members. —SS]

[Provide a plan to address the violations. This could include revising the triggers, if they are found to be too weak, strong or ambiguous. —SS]

10 Additional Productivity Metrics

[If your team has additional metrics of productivity, please feel free to add them to this report. —SS]